

Task 4: Report

Cucumber Disease Recognition

Introduction:

Our dataset is designed to train machine vision algorithms for the early recognition and classification of common cucumber diseases, addressing the challenge of farmers being unable to detect these issues with the naked eye.

Dataset Classes

The dataset comprises eight distinct classes of cucumber conditions:

- **Diseases (6 Classes):**
 1. Anthracnose
 2. Bacterial Wilt
 3. Belly Rot
 4. Downy Mildew
 5. Pythium Fruit Rot
 6. Gummy Stem Blight
- **Healthy Conditions (2 Classes):**
 1. Fresh leaves (presumably healthy leaves)
 2. Fresh cucumber (presumably healthy fruit)

Dataset Size and Structure

- Total Original Images: 1,280
- Images per Class: 160 images/class * 8 classes = 1,280 total images

Primary Goal

The primary goal of this entire study is the development and evaluation of robust Machine Vision algorithms for the early recognition and accurate classification of common cucumber diseases across eight distinct classes. The objective is to provide a reliable model capable of improving cucumber yields and utilization by enabling timely disease identification in agricultural settings.

Methodology

The evaluation process spanned three distinct phases, comparing five different Convolutional Neural Network (CNN) architectures based on their predictive performance and computational efficiency (training time).

Model Category	Architectures Evaluated	Core Mechanism
Fine-Tuned Pre-trained (Tasks 2)	ResNet50, VGG16, EfficientNetB0	Feature extraction from pre-trained ImageNet weights, training a custom classification head.
Custom CNN (Task 3)	Scratch-built architecture	Training a small, custom network from scratch without transfer learning.
Attention-Enhanced CNN (Task 4)	Attentive ResNet50 (CBAM)	Integration of the Convolutional Block Attention Module (CBAM) into the pre-trained ResNet50 base to refine feature focus.

All models were evaluated using a consistent test split, and comprehensive metrics including Accuracy, F1-score, Precision, Recall, Confusion Matrix, and ROC/AUC were calculated.

Results:

Performance Summary:

Model Name	Type	Attention Module	Test Accuracy	Weighted	Training Time (min)	ROC/AUC Computed
ResNet50 (Task 2/3)	Fine-Tuned	No	0.933	0.93	96.27	Yes
VGG16 (Task 2/3)	Fine-Tuned	No	0.897	0.89	203.51	Yes
EfficientNetB0 (Task 2/3)	Fine-Tuned	No	0.8299	0.8235	84.31	Yes
Custom CNN (Task 3)	Custom	No	0.7732	0.73	12.72	Yes

Model Name	Type	Attention Module	Test Accuracy	Weighted	Training Time (min)	ROC/AUC Computed
Attentive ResNet50 (CBAM)	Fine-Tuned	CBAM	0.89	0.89	19754.16	Yes

Performance Analysis

- **Top Performer:** The Fine-Tuned ResNet50 model achieved the highest predictive performance with an outstanding 93.3% Accuracy and 0.93 Weighted F1-score. This confirms that the rich, general-purpose features learned by ResNet50 on the massive ImageNet dataset are highly transferable and effective for this specific disease recognition task.
- **Custom Model Viability:** Custom CNN achieved a respectable 77.32% Accuracy, proving that a scratch-built model can perform adequately, though it lags behind all of the successfully fine-tuned pre-trained models.

Efficiency Analysis

- **Most Efficient:** Custom CNN is the most computationally efficient model, completing its training in a mere 12.72 minutes. This makes it an ideal choice for environments with severe computational resource constraints.
- **Least Efficient:** The VGG16 model was the slowest, requiring over three hours (203.51 minutes) for training, demonstrating the inefficiency of its older, deep architecture.

Conclusion

The evaluation establishes a clear trade-off between performance and efficiency across the models:

- **Recommendation for Accuracy:** The Fine-Tuned ResNet50 is the recommended model for deployment, offering the highest reliability (93.3% Accuracy). Its superior predictive capability is essential where misclassification of diseases carries a high cost.
- **Recommendation for Efficiency:** Custom CNN is the superior choice when training time and resource consumption are the primary constraints, providing a fast solution (12.72 min) with acceptable performance (77.32% Accuracy).

The attention mechanism, as implemented in this final task, failed to provide a performance boost, indicating that for this dataset, simple fine-tuning of robust pre-trained models is the most effective approach.